Rhetorical Synthesis — Part 3

Expression of Ideas · Reading and Writing · 68 questions

3.1 da67f89e-3c3e-4f3a-b7d2-21819515497c · Medium

While researching a topic, a student has taken the following notes:

- Planetary scientists classify asteroids based on their composition.
- C-type asteroids are composed primarily of carbon.
- They account for roughly 75 percent of known asteroids.
- S-type asteroids are primarily made up of silicate minerals.
- They account for roughly 17 percent of known asteroids.

The student wants to emphasize a difference between C-type and S-type asteroids. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Planetary scientists classify asteroids into types, two of which are the C-type and the S-type.

B. Planetary scientists consider an asteroid's composition (such as whether the asteroid is composed mainly of silicate minerals or carbon) when classifying it.

C. Roughly 17 percent of known asteroids are classified as S-type asteroids; another percentage is classified as C-type asteroids.

D. C-type asteroids are mainly composed of carbon, whereas S-type asteroids are primarily made up of silicate minerals.

3.2 b55400a0-38cf-4584-94ad-7d3b5cf2d570 · Medium

While researching a topic, a student has taken the following notes:

- Annie Wu is a prominent American flutist who graduated from the New England Conservatory.
- She has won multiple national flute competitions.
- She is best known for a 2011 YouTube video that has been viewed over two million times.
- The video shows her performing *Three Beats for Beatbox Flute,* an original work by composer Greg Pattillo.
- Wu combines flute playing and beatboxing in the video.

The student wants to emphasize Wu's most well-known achievement. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Annie Wu, who has won multiple national flute competitions, has also combined flute playing and beatboxing.

B. Among her many achievements, prominent American flutist Annie Wu graduated from the New England Conservatory and has won multiple national flute competitions.

C. Annie Wu is best known for a 2011 YouTube video performance of *Three Beats for Beatbox Flute* that has been viewed over two million times.

D. Composer Greg Pattillo's original work *Three Beats for Beatbox Flute* combines flute playing and beatboxing.

3.3 bdd1b170-ff13-4936-a7ea-0f1ff0845133 · Medium

While researching a topic, a student has taken the following notes:

- In astronomy, the mass of stars can be described in units called solar masses.
- One solar mass is roughly equal to the mass of the Sun.
- The mass of the star Proxima Centauri is 0.122 solar masses.
- The mass of the star Sirius A is 2.063 solar masses.

The student wants to emphasize the mass of Sirius A. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. The mass of stars, like Proxima Centauri, can be described in units called solar masses.

B. In astronomy, the mass of stars can be described in units called solar masses, and one solar mass is roughly equal to the mass of the Sun.

C. The Sun is more massive than Proxima Centauri, which has a mass of 0.122 solar masses.

D. With a mass of 2.063 solar masses, Sirius A is more massive than the Sun.

3.4 d6d5fa78-74b9-4d37-a15f-af9162d592aa · Medium

While researching a topic, a student has taken the following notes:

- Pinnipeds, which include seals, sea lions, and walruses, live in and around water.
- Pinnipeds are descended not from sea animals but from four-legged, land-dwelling carnivores.
- Canadian paleobiologist Natalia Rybczynski recently found a fossil with four legs, webbed toes, and the skull and teeth of a seal.
- Rybczynski refers to her rare find as a “transitional fossil.”
- The fossil illustrates an early stage in the evolution of pinnipeds from their land-dwelling ancestors.

The student wants to emphasize the fossil’s significance. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Canadian paleobiologist Natalia Rybczynski’s fossil has the skull and teeth of a seal, which, like sea lions and walruses, is a pinniped.

B. Pinnipeds are descended from four-legged, land-dwelling carnivores; a fossil that resembles both was recently found.

C. Having four legs but the skull and teeth of a seal, the rare fossil illustrates an early stage in the evolution of pinnipeds from their land-dwelling ancestors.

D. A “transitional fossil” was recently found by paleobiologist Natalia Rybczynski.

3.5 3be7c190-7615-4bc8-b801-6780fee56ba0 · Medium

While researching a topic, a student has taken the following notes:

- Stars form in a galaxy when gravity causes a massive cloud of dust and gas to collapse.
- A galaxy in a phase of rapid star formation is called a starburst galaxy.
- Quenching is a process in which a galaxy loses star-forming gas.
- A galaxy that no longer forms stars is called a quenched galaxy.
- A quenched galaxy has entered the poststarburst phase.

The student wants to explain what a quenched galaxy is. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Before quenching, a starburst galaxy will form stars at a rapid rate.

B. When it becomes quenched, a starburst galaxy enters the poststarburst phase.

C. Having entered the poststarburst phase, a quenched galaxy is one that no longer forms stars.

D. A starburst galaxy will lose star-forming gas and eventually become quenched.

3.6 bdfb2b51-a55e-4114-a6fe-e6b6206c8b3c · Medium

While researching a topic, a student has taken the following notes:

- Muckrakers were journalists who sought to expose corruption in US institutions during the Progressive Era (1897–1920).
- Ida Tarbell was a muckraker who investigated the Standard Oil Company.
- She interviewed Standard Oil Company executives, oil industry workers, and public officials.
- She examined thousands of pages of the company's internal communications, including letters and financial records.
- Her book *The History of the Standard Oil Company* (1904) exposed the company's unfair business practices.

The student wants to emphasize the thoroughness of Ida Tarbell's investigation of the Standard Oil Company. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Ida Tarbell not only interviewed Standard Oil executives, oil industry workers, and public officials but also examined thousands of pages of the company's internal communications.

B. Ida Tarbell, who investigated the Standard Oil Company, was a muckraker (a journalist who sought to expose corruption in US institutions during the Progressive Era, 1897–1920).

C. As part of her investigation of the Standard Oil Company, muckraker Ida Tarbell conducted interviews.

D. Published in 1904, muckraker Ida Tarbell's book *The History of the Standard Oil Company* exposed the company's unfair business practices.

3.7 4404a2b9-0e73-42a1-9d83-4f0019e197aa · Medium

While researching a topic, a student has taken the following notes:

- When medical students mention their patients on social media, they may violate patient confidentiality.
- Terry Kind led a study to determine how many medical schools have student policies that mention social media use.
- Kind and her team reviewed 132 medical school websites, examining publicly available student policies.
- Only thirteen medical schools had guidelines that explicitly mention social media, and only five defined what constitutes acceptable social media use.

The student wants to emphasize the study's methodology. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. The student policies of 132 medical schools can be found online, according to research by Terry Kind.

B. To find out how many medical schools have guidelines about student social media use, Terry Kind and her team examined the student policies of 132 medical schools.

C. Out of 132 medical schools, only thirteen had student policies that mentioned social media, and only five specified what use was acceptable.

D. Terry Kind and her team wanted to know how many medical schools have student social media policies in place about protecting patient confidentiality.

3.8 6f77f810-8fba-4625-b6e7-411afe24b747 · Medium

While researching a topic, a student has taken the following notes:

- Pterosaurs were flying reptiles that existed millions of years ago.
- In a 2021 study, Anusuya Chinsamy-Turan analyzed fragments of pterosaur jawbones located in the Sahara Desert.
- She was initially unsure if the bones belonged to juvenile or adult pterosaurs.
- She used advanced microscope techniques to determine that the bones had few growth lines relative to the bones of fully grown pterosaurs.
- She concluded that the bones belonged to juveniles.

The student wants to present the study and its findings. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. In 2021, Chinsamy-Turan studied pterosaur jawbones and was initially unsure if the bones belonged to juveniles or adults.

B. Pterosaur jawbones located in the Sahara Desert were the focus of a 2021 study.

C. In a 2021 study, Chinsamy-Turan used advanced microscope techniques to analyze the jawbones of pterosaurs, flying reptiles that existed millions of years ago.

D. In a 2021 study, Chinsamy-Turan determined that pterosaur jawbones located in the Sahara Desert had few growth lines relative to the bones of fully grown pterosaurs and thus belonged to juveniles.

3.9 fc018601-4465-4f04-82ee-06f6db2cf75b · Medium

While researching a topic, a student has taken the following notes:

- Traditionally, manufacturers have dyed denim jeans blue by dipping them in a solution containing indigo powder.
- Indigo doesn't dissolve in just water, so manufacturers must mix hazardous chemicals with water to dissolve the powder.
- Textile researcher Smriti Rai discovered a process for dyeing blue jeans without chemicals.
- Rai added indigo powder to a hydrogel containing nanocellulose and produced a dye that could be spread directly onto the denim.
- Nanocellulose is a natural, plant-based substance that separates the molecules of indigo powder.

The student wants to emphasize a difference between the two approaches to dyeing blue jeans. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Though created using a different process, Rai's dye contains the same ingredient as the dye produced by blue jean manufacturers.

B. Nanocellulose is a natural, plant-based substance that separates the molecules of indigo powder, which doesn't dissolve in water.

C. The traditional approach to dyeing blue jeans is to dip them in a solution containing hazardous chemicals.

D. Rai's approach substitutes a natural, plant-based substance for the hazardous chemicals that manufacturers have traditionally used.

3.10 5a4e75f4-a147-4de8-a439-6025b2d19bb0 · Medium

While researching a topic, a student has taken the following notes:

- Iranian scholar Abu Rayhan al-Biruni studied Earth's physical features.
- He theorized that a large landmass existed west of Europe and east of Asia.
- Al-Biruni published his landmass theory in 1037 CE.

The student wants to specify when al-Biruni published his landmass theory. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. In 1037 CE, al-Biruni published his theory that a large landmass existed west of Europe and east of Asia.

B. Al-Biruni, who studied Earth's physical features, published a theory about a large landmass.

C. Al-Biruni was an Iranian scholar who studied Earth's physical features.

D. An Iranian scholar who studied Earth's physical features, al-Biruni theorized that a large landmass existed west of Europe and east of Asia.

3.11 bbc4e5f1-ff38-4b85-a2b3-9bae4f358a79 · Medium

While researching a topic, a student has taken the following notes:

- The melting rate of glaciers varies based on air temperature.
- In the warm summer months, massive glaciers on the coast of Greenland melt into the surrounding water.
- The melting glaciers contribute to rising sea levels each summer.
- Huge icebergs also break off Greenland's glaciers into the water and melt.
- In 2017, geoscientist Twila Moon found that the iceberg melting rate depends not on air temperature but on water temperature.
- Because water temperature is consistent, melting icebergs contribute to rising sea levels all year.

The student wants to emphasize a similarity between glaciers and icebergs in Greenland. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Because icebergs break off Greenland's glaciers into the water, their melting rate depends on water temperature.

B. Greenland's glaciers and icebergs both melt during the year, contributing to rising sea levels.

C. Geoscientist Twila Moon found that the melting rate of Greenland's icebergs, unlike that of glaciers, does not depend on air temperature.

D. Glaciers on the coast of Greenland melt during the warm summer months into the surrounding water, the temperature of which remains consistent throughout the year.

3.12 0b406856-4131-43d4-a343-4d2db939e35e · Medium

While researching a topic, a student has taken the following notes:

- 1914: British explorer Ernest Shackleton and a small crew embarked on an expedition to Antarctica.
- 1915: Shackleton's ship *Endurance* became stuck in ice before eventually breaking apart and sinking.
- 1916: After more harrowing sea-ice adventures, the entire crew was rescued.
- 1959: Historian Alfred Lansing wrote a book called *Endurance: Shackleton's Incredible Voyage.*
- 2001: Filmmaker George Butler released a documentary called *The Endurance: Shackleton's Legendary Antarctic Expedition*.
- 2022: The wreckage of *Endurance* was discovered at the bottom of Antarctica's Weddell Sea.

The student wants to provide a historical overview of the Shackleton expedition. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Leaving in 1914 for Antarctica, Shackleton and his crew underwent many harrowing sea-ice adventures, including losing their ship in 1915, before being rescued in 1916.

B. In 1914, the Shackleton expedition sailed to Antarctica, where, in 1916, they rescued the crew of a ship that had sunk, *Endurance* (the wreckage of which was discovered in 2022).

C. Shackleton's expedition has inspired a 1959 book, a 2001 film, and a 2022 discovery.

D. Alfred Lansing wrote about the history of Shackleton's 1914–16 expedition in the book *Endurance: Shackleton's Incredible Voyage* (1959); years later, in 2001, George Butler released a documentary about the expedition.

3.13 295d509c-ccd0-4431-a2f8-9a68b59825c6 · Medium

While researching a topic, a student has taken the following notes:

- The Pueblo of Zuni is located about 150 miles west of Albuquerque, New Mexico.
- It is the traditional home of the A:shiwi (Zuni) people.
- The A:shiwi A:wan Museum and Heritage Center was established by tribal members in 1992.
- Its mission is stated on its website: “As a tribal museum and heritage center for the Zuni people and by the Zuni people we work to provide learning experiences that emphasize A:shiwi ways of knowing, as well as exploring modern concepts of knowledge and the transfer of knowledge.”

The student wants to emphasize how long the museum has existed. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. The Pueblo of Zuni is home to the A:shiwi A:wan Museum and Heritage Center, which was founded by tribal members.

B. The A:shiwi A:wan Museum and Heritage Center has served the Pueblo of Zuni since 1992.

C. According to its website, the A:shiwi A:wan Museum and Heritage Center (founded in the 1990s) works to “emphasize A:shiwi ways of knowing.”

D. Knowledge has been one of the central themes of the A:shiwi A:wan Museum and Heritage Center from its founding.

3.14 8e115ea7-bbb9-40f5-b9fe-5a8745b61654 · Medium

While researching a topic, a student has taken the following notes:

- The meter of a poem is the rhythmic structure or pattern of accents in its lines.
- Alliterative meter is structured by a pattern of repeated sounds.
- Quantitative meter is structured by a pattern of long and short syllables.
- The Old English poem *Widsith* uses an alliterative meter.
- The Sanskrit poem *Meghadūta* uses a quantitative meter.

The student wants to emphasize a difference between the meters of the two poems. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. The poem *Widsith* is written in Old English, but *Meghadūta* is written in Sanskrit.

B. Alliterative meter is a pattern of repeated sounds, but quantitative meter is a rhythmic structure or pattern of accents.

C. The Sanskrit poem *Meghadūta* uses a quantitative meter, while the meter of the Old English poem *Widsith* uses a pattern of long and short syllables.

D. The lines of the poem *Meghadūta* use a pattern of long and short syllables, whereas *Widsith*'s lines use a pattern of repeated sounds.

3.15 a0186a2e-4208-4287-b388-b90a267672ef · Medium

While researching a topic, a student has taken the following notes:

- Just like states have state flags, some cities have city flags.
- Over one hundred US cities have redesigned their flags since 2015.
- The city of Pocatello, Idaho, redesigned its flag after it was named the most poorly designed flag in North America.
- Pocatello's new flag better represents the city's mountainous geography and civic priorities.
- Residents consider the new flag to be a meaningful symbol of civic pride.

The student wants to make and support a generalization about the effect of redesigning a city flag. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Over one hundred US cities have redesigned their flags, including Pocatello, whose flag had been named the most poorly designed flag in North America.

B. Pocatello is just one of over one hundred US cities that have redesigned their flags.

C. After it was named the most poorly designed flag in North America, the flag of Pocatello was redesigned to better represent the city's geography and civic priorities.

D. Redesigning a poorly designed city flag can create a meaningful symbol of civic pride, as was the case when Pocatello redesigned its original flag to better represent its geography and civic priorities.

3.16 e934e3ce-75e8-4e0c-b825-5e84358e4638 · Medium

While researching a topic, a student has taken the following notes:

- Jordan Bennett is a Mi'Kmaq visual artist.
- The Mi'Kmaq are a First Nations people in North America.
- Bennett's paintings pay homage to traditional Mi'Kmaq craftsmanship and have been displayed in over 75 exhibitions.
- His 2017 exhibition *Wije'wi* was held at the Grenfell Art Gallery.
- His 2018 exhibition *Ketu'elmita'jik* was held at the Art Gallery of Nova Scotia.

The student wants to emphasize the order in which two of Jordan Bennett's exhibitions were held. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Jordan Bennett's 2017 exhibition *Wije'wi* was followed a year later by his exhibition *Ketu'elmita'jik.*

B. Jordan Bennett's paintings, some of which appeared in 2017 and 2018 exhibitions, pay homage to traditional Mi'Kmaq craftsmanship.

C. Mi'Kmaq visual artist Jordan Bennett has displayed his work in over 75 exhibitions, including *Wije'wi* and *Ketu'elmita'jik.*

D. Jordan Bennett's 2018 exhibition *Ketu'elmita'jik* was held at the Art Gallery of Nova Scotia; another was held at the Grenfell Art Gallery.

3.17 e9f6a4dd-516c-4460-a847-a065224c7b93 · Medium

While researching a topic, a student has taken the following notes:

- Little is known about the life of Wong Fei-hung (1847–1925).
- He was born near Foshan, China, and gained local recognition as a physician and Hung Ga (also known as Hung Gar) Kung Fu master.
- He achieved many incredible martial arts feats—some confirmed and some rumored.
- He has become an internationally known folk hero thanks to his depiction in over a hundred films, television shows, and other media.
- In the 1991 film *Once Upon a Time in China*, actor Jet Li portrays Wong Fei-hung using superhuman kung fu abilities to save his community.

The student wants to emphasize the effect media had on building Wong Fei-hung's legacy. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Thanks to his depiction in over a hundred pieces of media, Wong Fei-hung was locally known as a successful physician and Hung Ga Kung Fu master.

B. Though he was known locally during his lifetime, Wong Fei-hung's later depiction in television, film, and other media has turned him into an internationally known folk hero.

C. Various media have depicted Wong Fei-hung, the successful physician and kung fu master who became an internationally known folk hero.

D. Wong Fei-hung's abilities as a kung fu master are depicted in many media, including the 1991 film *Once Upon a Time in China*.

3.18 ba326bd3-4344-46af-9257-c70f53076429 · Medium

While researching a topic, a student has taken the following notes:

- In 2013, paleontology professor Hesham Sallam and his students from Mansoura University in Egypt made a discovery.
- The team found a partial dinosaur skeleton at a site in Egypt's Dakhla Oasis.
- The skeleton belonged to a dinosaur species that lived approximately 80 million years ago.
- The new species was named *Mansourasaurus* to recognize the team that discovered it.

The student wants to explain the origin of the species' name. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. *Mansourasaurus,* a new species discovered in Egypt in 2013, lived approximately 80 million years ago.

B. A partial dinosaur skeleton found in Egypt's Dakhla Oasis belonged to a species named *Mansourasaurus*.

C. *Mansourasaurus,* a species that lived approximately 80 million years ago, was discovered in 2013 by Egyptian paleontologist Hesham Sallam and a team of university students.

D. The new species was named *Mansourasaurus* to recognize the team that discovered it, a professor and students from Mansoura University.

3.19 c3fb024d-054c-423f-bb42-cf33987d4d10 · Medium

While researching a topic, a student has taken the following notes:

- If a moon orbiting a planet comes close enough to that planet, tidal forces can cause the moon to break apart.
- In a 2022 study, researchers proposed that Saturn was once orbited by a large moon they named Chrysalis.
- Their simulations indicated that Chrysalis would likely have come very close to Saturn around 160 million years ago.
- At that distance, Chrysalis would have been broken apart by tidal forces.
- The researchers hypothesized that the resulting debris formed Saturn's rings.

The student wants to recount the sequence of events proposed by the researchers. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. According to researchers' simulations, two events likely occurred around 160 million years ago: first, Chrysalis came very close to Saturn, and second, debris from Saturn's rings caused the moon to break apart.

B. If a moon orbiting a planet (like Saturn) comes close enough to that planet, tidal forces can cause the moon to break apart.

C. Around 160 million years ago, a large moon (Chrysalis) came close enough to Saturn that tidal forces broke the moon apart; its debris then formed the planet's rings.

D. First, researchers proposed that Saturn was orbited by a large moon (Chrysalis); next, they conducted simulations; and, finally, they formed a hypothesis.

3.20 34eafb3e-bcab-4b7b-b488-987611a37efb · Medium

While researching a topic, a student has taken the following notes:

- Gaspar Enriquez is an artist.
- He specializes in portraits of Mexican Americans.
- A portrait is an artistic representation of a person.
- Enriquez completed a painting of the sculptor Luis Jimenez in 2003.
- He completed a drawing of the writer Rudolfo Anaya in 2016.

The student wants to emphasize a difference between the two portraits. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. The portraits, or artistic representations, of Luis Jimenez and Rudolfo Anaya were both completed by Enriquez in the early 2000s.

B. Enriquez has completed portraits of numerous Mexican Americans, including sculptor Luis Jimenez and writer Rudolfo Anaya.

C. While both are by Enriquez, the 2003 portrait of Luis Jimenez is a painting, and the 2016 portrait of Rudolfo Anaya is a drawing.

D. Luis Jimenez was a Mexican American sculptor, and Rudolfo Anaya was a Mexican American writer.

3.21 44184b07-d595-4b34-bf13-95e1d824be36 · Medium

While researching a topic, a student has taken the following notes:

- In a 2020 study, researchers in California investigated how many potential nesting sites female wood ducks visited during the nesting season.
- The researchers placed nest boxes throughout the survey area and tagged 138 female wood ducks with radio frequency ID trackers.
- These trackers recorded how many nest boxes each duck visited.
- 67 ducks (48.5%) visited only one nest box.
- 18 ducks (13.0%) visited 10 or more nest boxes.
- Younger ducks were more likely to visit multiple nest boxes.

The student wants to present the methods used in the 2020 study. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. By recording how many nest boxes each duck visited, the researchers discovered that only a relatively small percentage (13.0%) of the ducks visited 10 or more nest boxes.

B. After tracking how many nest boxes the 138 wood ducks visited, the researchers found that the younger ducks tended to visit more nest boxes than the older ducks.

C. The researchers tagged 138 female wood ducks with radio frequency ID trackers and recorded how many nest boxes each duck visited during the nesting season.

D. The researchers investigated each nesting site for signs that it had been visited by the female wood ducks.

3.22 977bf489-037c-4bce-bcb0-1dc735c870e4 · Medium

While researching a topic, a student has taken the following notes:

- Archaeologist Dr. Sada Mire founded the Horn Heritage Foundation to preserve the cultural history of regions in the Horn of Africa.
- Horn Heritage has overseen a preservation project to create 3D digital scans of ancient rock art in Somaliland.
- Paintings found at the Laas Geel caves are included in the scans.
- The Laas Geel paintings feature human figures and animals.
- Paintings found at the Dhagah Nabi Galay caves are included in the scans.
- The Dhagah Nabi Galay caves feature what are thought to be the earliest examples of writing in East Africa.

The student wants to emphasize a similarity between the Laas Geel paintings and the Dhagah Nabi Galay paintings. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. The earliest examples of writing in East Africa are thought to be featured in the paintings at the Dhagah Nabi Galay caves in Somaliland.

B. The paintings at the Dhagah Nabi Galay caves feature examples of writing, while those at the Laas Geel caves feature humans and animals.

C. In Somaliland, the paintings in the Laas Geel caves feature human figures and animals.

D. The Laas Geel paintings and the Dhagah Nabi Galay paintings are both examples of ancient rock art found in Somaliland.

3.23 e1a88e15-d0ab-4fb0-8ed9-5e5f24396021 · Medium

While researching a topic, a student has taken the following notes:

- The popular wood-wide web theory posits that trees can communicate and exchange resources with one another via common mycorrhizal networks (CMNs) of fungi.
- Ecologist Dr. Suzanne Simard first suggested this theory in 1997.
- She described trees as “super-cooperators.”
- In the 2022 study “The Decay of the Wood-Wide Web?,” mycologist Dr. Justine Karst and colleagues evaluated dozens of CMN studies.
- They write that CMNs “have captured the interest of broad audiences. We are concerned, however, that recent claims about CMNs in forests are disconnected from evidence.”

The student wants to use a quotation to emphasize a potential problem with the wood-wide web theory. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Describing trees as “super-cooperators,” Simard first suggested that trees can exchange resources with one another in 1997.

B. In “The Decay of the Wood-Wide Web?,” Karst and colleagues note that common mycorrhizal networks “have captured the interest of broad audiences.”

C. After evaluating dozens of CMN studies, Karst and colleagues expressed concern that recent claims about common mycorrhizal networks are “disconnected from evidence.”

D. Despite the concerns expressed in the 2022 study “The Decay of the Wood-Wide Web?,” the wood-wide web theory remains popular.

3.24 31f36559-8bb0-4c98-b21a-c714433d07f2 · Medium

While researching a topic, a student has taken the following notes:

- Crown shyness is a phenomenon in which the tops (crowns) of neighboring trees grow close together but don't overlap.
- To explain how this happens, Australian forester M.R. Jacobs proposes the mutual abrasion theory.
- According to Jacobs's theory, when trees brush against one another, branches break off.
- Malaysian scholar Francis S.P. Ng posits the mutual shade avoidance theory.
- According to Ng's theory, when tree branches detect shade from nearby trees' branches, they stop growing.

The student wants to compare the causes of crown shyness proposed in the two theories. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. While Jacobs proposes that crown shyness is caused by neighboring tree branches brushing against one another, Ng posits that it occurs when branches detect shade from nearby trees' branches.

B. Both Jacobs and Ng have proposed theories to explain what causes crown shyness.

C. Ng posits the mutual shade avoidance theory, whereas Jacobs proposes an alternative theory.

D. Jacobs's mutual abrasion theory proposes that when neighboring trees brush against one another, branches break off, resulting in a phenomenon in which the tops of trees grow close together but don't overlap.

3.25 75e7d111-ced9-4da0-9be1-0c9cd8da1f3b · Medium

While researching a topic, a student has taken the following notes:

- A wok is a cooking pan that originated in China during the Han dynasty (206 BCE–220 CE).
- The wok's round, wide base helps to cook food evenly.
- The wok's high, angled sides help to contain oil splatters.
- Grace Young is a cook and culinary historian.
- Her book *The Breath of a Wok* (2004) traces the history of the wok.

The student wants to describe the wok's shape. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Grace Young's 2004 book, *The Breath of a Wok*, traces the history of the cooking pan.

B. Able to cook food evenly and contain oil splatters, the wok is the subject of Grace Young's 2004 book.

C. A wok is a cooking pan with a round, wide base and high, angled sides.

D. The design of a wok, a type of cooking pan that originated in China during the Han dynasty, helps the pan cook food evenly and contain oil splatters.

3.26 55b902e4-fa91-44f6-8109-7dc4272cd236 · Hard

While researching a topic, a student has taken the following notes:

- Astronomers estimate that the number of comets orbiting the Sun is in the billions.
- 81P/Wild is one of many comets whose orbit has changed over time.
- 81P/Wild's orbit once lay between the orbits of Uranus and Jupiter.
- The comet's orbit is now positioned between the orbits of Jupiter and Mars.

The student wants to make and support a generalization about the orbits of comets. Which choice most effectively uses relevant information from the notes to accomplish these goals?

A. Astronomers estimate that the number of comets orbiting the Sun is in the billions; the comets' orbits may change over time.

B. Like Uranus, Jupiter, and Mars, billions of comets orbit the Sun.

C. One example of a comet is 81P/Wild, whose orbit around the Sun once lay between Uranus's and Jupiter's orbits but is now positioned between those of Jupiter and Mars.

D. A comet's orbit around the Sun may change over time: the orbit of comet 81P/Wild once lay between the orbits of Uranus and Jupiter but is now positioned between those of Jupiter and Mars.

3.27 d245bdf2-d209-40aa-a94a-fd9d8eac1fd6 · Hard

While researching a topic, a student has taken the following notes:

- The Gullah are a group of African Americans who have lived in parts of the southeastern United States since the 18th century.
- Gullah culture is influenced by West African and Central African traditions.
- Louise Miller Cohen is a Gullah historian, storyteller, and preservationist.
- She founded the Gullah Museum of Hilton Head Island, South Carolina, in 2003.
- Vermelle Rodrigues is a Gullah historian, artist, and preservationist.
- She founded the Gullah Museum of Georgetown, South Carolina, in 2003.

The student wants to emphasize the duration and purpose of Cohen's and Rodrigues's work. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. At the Gullah Museums in Hilton Head Island and Georgetown, South Carolina, visitors can learn more about the Gullah people who have lived in the region for centuries.

B. Louise Miller Cohen and Vermelle Rodrigues have worked to preserve the culture of the Gullah people, who have lived in the United States since the 18th century.

C. Since 2003, Louise Miller Cohen and Vermelle Rodrigues have worked to preserve Gullah culture through their museums.

D. Influenced by the traditions of West and Central Africa, Gullah culture developed in parts of the southeastern United States in the 18th century.

3.28 5f527608-5cc1-4525-bcc0-e04a71f57baf · Hard

While researching a topic, a student has taken the following notes:

- Leaders of the Province of Guatemala proclaimed independence for Central America from the Spanish Empire on September 15, 1821.
- The accompanying Declaration of Independence was written by Honduran scholar and politician José Cecilio del Valle.
- The 1812 Spanish Constitution had provided some degree of independence for Central America, but it was repealed by the Spanish king in 1814.
- Valle, a loyal advisor to the Spanish Empire's administrators in Central America, had long opposed independence.
- He changed his mind after Colonel Rafael del Riego's 1820 revolt, which demanded the return of rights lost in 1814.

The student wants to place the 1821 Declaration of Independence in the context of Valle's changing political beliefs. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Colonel Riego's revolt was the inspiration that Valle, a long-standing opponent of Central American independence, needed to change his political beliefs.

B. Long an opponent of Central American independence, Valle changed his mind after an 1820 revolt and wrote the 1821 declaration.

C. A change in Valle's political beliefs that occurred when the Spanish king repealed the 1812 constitution led to Valle writing Central America's Declaration of Independence.

D. The writing of Central America's Declaration of Independence may not have happened were it not for Colonel Riego's 1820 revolt.

3.29 91d983a0-c121-4eb2-a3cf-ee53649fe5aa · Hard

While researching a topic, a student has taken the following notes:

- British musicians John Lennon and Paul McCartney shared writing credit for numerous Beatles songs.
- Many Lennon-McCartney songs were actually written by either Lennon or McCartney, not by both.
- The exact authorship of specific parts of many Beatles songs, such as the verse for “In My Life,” is disputed.
- Mark Glickman, Jason Brown, and Ryan Song used statistical methods to analyze the musical content of Beatles songs.
- They concluded that there is 18.9% probability that McCartney wrote the verse for “In My Life,” stating that the verse is “consistent with Lennon’s songwriting style.”

The student wants to make a generalization about the kind of study conducted by Glickman, Brown, and Song. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Based on statistical analysis, Glickman, Brown, and Song claim that John Lennon wrote the verse of “In My Life.”

B. There is only an 18.9% probability that Paul McCartney wrote the verse for “In My Life”; John Lennon is the more likely author.

C. It is likely that John Lennon, not Paul McCartney, wrote the verse for “In My Life.”

D. Researchers have used statistical methods to address questions of authorship within the field of music.

3.30 459668c5-8725-4aa8-8529-d73621d54e4c · Hard

While researching a topic, a student has taken the following notes:

- The ancient Arab dhow was a sailing vessel distinguishable by its triangular sails and stitched hull construction.
- Dhows were used primarily for trade along the coasts of Arab, South Asian, and East African countries.
- Contemporary shipbuilders in Oman use a mix of modern and traditional materials to build replicas of ancient dhows.
- Most of the materials used are traditional.
- Replica hulls are stitched together using the same traditional coconut palm fiber rope used on the hulls of ancient dhows.

The student wants to make a generalization about the materials used in dhow replicas. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. A traditional material that was used to stitch together the hulls of ancient dhows, coconut palm fiber rope is still used by shipbuilders.

B. The ancient Arab dhow was a sailing vessel used primarily for trade and distinguishable by its triangular sails.

C. Although most materials used in dhow replicas are traditional, some modern materials are used.

D. Contemporary shipbuilders in Oman build replicas of the dhow, which was an ancient sailing vessel with a stitched hull construction.

3.31 737599e7-dea7-4978-87c8-0ddf52eecc44 · Hard

While researching a topic, a student has taken the following notes:

- Nissologists are scientists who study islands.
- Some nissologists define an island as any piece of land surrounded by water.
- Using that definition, they determined that Sweden has 221,000 islands.
- Other nissologists define an island as being 1 kilometer square, a certain distance from the mainland, and having at least 50 permanent residents.
- Using that definition, they determined that Sweden has 24 islands.

The student wants to make and support a generalization about nissologists' definition of an island. Which choice most effectively uses relevant information from the notes to accomplish these goals?

A. The definition of an island as any piece of land surrounded by water is supported by some nissologists, scientists who study islands.

B. Multiple counts of Sweden's islands have been based on different definitions of an island.

C. Based on a recent count, Sweden has a relatively small number of islands with at least 50 permanent residents.

D. Nissologists' different definitions can result in huge disparities in counts of islands, as the example of Sweden shows.

3.32 69bb5669-cf08-40db-a385-8a5074a1920c · Hard

While researching a topic, a student has taken the following notes:

- British scholar Robert Plot described fossilized dinosaur bones in his 1676 book *The Natural History of Oxfordshire.*
- Plot earned a reputation for being the first person to have discovered dinosaur remains.
- In 1990, archaeologists in Lesotho, in southern Africa, discovered a fossilized phalanx of a *Massospondylus carinatus* dinosaur in a cave once inhabited by humans.
- Indigenous Khoesan and Basotho peoples had inhabited the cave beginning around 1100 CE.
- According to paleontologist Julien Benoit, these peoples may have found the phalanx and brought it to the cave centuries before Plot's descriptions.

The student wants to emphasize the significance of the 1990 discovery to Plot's reputation. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Benoit challenged Plot's reputation for being the first person to have discovered *M. carinatus* remains.

B. Evidence that Khoesan and Basotho peoples may have found an *M. carinatus* phalanx as long ago as 1100 CE suggests that Plot may not have been the first person to have discovered dinosaur remains.

C. According to Benoit's analysis of the 1990 discovery, Indigenous peoples in southern Africa may have brought the fossilized phalanx to the cave as long ago as 1100 CE.

D. In 1990, more than three centuries after Plot claimed in his book that he had found fossilized dinosaur bones, archaeologists uncovered evidence in southern Africa that disproved his claims.

3.33 b587af2d-a137-4b31-a984-05d6e631b3ed · Hard

While researching a topic, a student has taken the following notes:

- Archaeologist Jon Erlandson and colleagues argue that humans first arrived in the Americas by sea.
- They propose that humans traveled between Pacific Ocean islands and coastlines from northeast Asia to the Americas.
- Many of these islands and coastal zones were later submerged as glaciers melted and sea levels rose.
- The researchers think that “a coastal route, including kelp forests and estuaries, would have provided a rich mix of marine, estuarine, riverine, and terrestrial resources” such as seaweeds, fish, and birds.
- This proposed scenario is known as the kelp highway hypothesis.

The student wants to summarize the kelp highway hypothesis. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Pacific Ocean islands and coastlines likely contained “a rich mix of marine, estuarine, riverine, and terrestrial resources” such as seaweeds, fish, and birds, according to researchers.

B. One argument about how humans first arrived in the Americas is the kelp highway hypothesis proposed by Jon Erlandson and colleagues.

C. Humans may have first arrived in the Americas by sea, traveling between Pacific Ocean islands and coastlines and subsisting on a variety of resources.

D. As glaciers melted and sea levels rose, many Pacific Ocean islands and coastal zones were submerged.

3.34 f0ebdaaf-0942-435d-9436-45fbbbf064ee · Hard

While researching a topic, a student has taken the following notes:

- Ulaanbaatar is the capital of Mongolia.
- The city's population is 907,802.
- Ulaanbaatar contains 31.98 percent of Mongolia's population.
- Hanoi is the capital of Vietnam.
- The city's population is 7,781,631.
- Hanoi contains 8.14 percent of Vietnam's population.

The student wants to emphasize the relative sizes of the two capitals' populations. Which choice most effectively uses information from the given sentences to emphasize the relative sizes of the two capitals' populations?

A. Mongolia's capital is Ulaanbaatar, which has 907,802 people, and Vietnam's capital is Hanoi, which has 7,781,631 people.

B. Comparing Vietnam and Mongolia, 7,781,631 is 8.14 percent of Vietnam's population, and 907,802 is 31.98 percent of Mongolia's.

C. Even though Hanoi (population 7,781,631) is larger than Ulaanbaatar (population 907,802), Ulaanbaatar accounts for more of its country's population.

D. The populations of the capitals of Mongolia and Vietnam are 907,802 (Ulaanbaatar) and 7,781,631 (Hanoi), respectively.

3.35 8a56d1a4-d07b-451f-b4f9-1b644a8f0a5b · Hard

While researching a topic, a student has taken the following notes:

- The fifth Solvay Conference on Physics was held in 1927.
- It brought together twenty-nine of the era's preeminent scientists to discuss the emerging field of quantum theory.
- The conference famously featured a debate between physicists Albert Einstein and Niels Bohr.
- Bohr proposed that subatomic entities like electrons had only probable realities until they were observed.
- Einstein argued that subatomic entities like electrons had a reality independent of observation.
- Bohr's position, later called the Copenhagen interpretation, remains the most widely accepted theory of quantum mechanics.

The student wants to place Einstein's argument within its historical context. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. During the dawn of quantum theory, Einstein maintained the independent reality of some subatomic entities, although Bohr's opposing interpretation would become the widely accepted view.

B. At the 1927 Solvay Conference on Physics, Einstein disagreed with Bohr's argument that subatomic entities like electrons had a reality independent of observation.

C. The attendees of the 1927 Solvay Conference were among the preeminent scientists of their era, including Einstein, who opposed Bohr's proposal.

D. In 1927, Einstein and Bohr engaged in a famous debate; Bohr's argument, later called the Copenhagen interpretation, would remain popular decades after.

3.36 b6d902ea-2453-43ee-99d4-d234a13bfa07 · Hard

While researching a topic, a student has taken the following notes:

- The farm-size transition hypothesis predicts that economic pressures associated with modernization result in smaller farms amalgamating into larger-scale commercial farms.
- Masters et al. (2013): The average farm size in Asia "already has or will soon begin to rise."
- Promkhambut et al. (2023) argue that small rice farms in Thailand have adopted modern farming methods without a significant scaling-up of farm size.
- Promkhambut et al.: "The persistence of [small] rice farms [in Thailand] does not represent a 'failure' to modernize...or a 'truncated' transition—it is a *response* to modernization."

The student wants to make and support a claim regarding the applicability of the farm-size transition hypothesis to Thailand. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Taken together, the studies by Masters et al. and Promkhambut et al. suggest that rice farms in Thailand have responded to the economic pressures associated with modernization by expanding in size.

B. Masters et al. report that the average farm size "already has or will soon begin to rise" in Asia, a finding that is consistent with the farm-size transition hypothesis.

C. The predicted shift to large-scale commercial farming may not hold true for rice farms in Thailand, where, according to Promkhambut et al., rice farms have remained small as they've modernized.

D. Although the farm-size transition hypothesis may be applicable to some countries in Asia, it is inconsistent with the development of rice farming in Thailand.

3.37 bc609dfd-b00c-4820-b30b-0bb483974080 · Hard

While researching a topic, a student has taken the following notes:

- Ducklings expend up to 62.8% less energy when swimming in a line behind their mother than when swimming alone.
- The physics behind this energy savings hasn't always been well understood.
- Naval architect Zhiming Yuan used computer simulations to study the effect of the mother duck's wake.
- The study revealed that ducklings are pushed in a forward direction by the wake's waves.
- Yuan determined this push reduces the effect of wave drag on the ducklings by 158%.

The student wants to present the study and its methodology. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. A study revealed that ducklings, which expend up to 62.8% less energy when swimming in a line behind their mother, also experience 158% less drag.

B. Seeking to understand how ducklings swimming in a line behind their mother save energy, Zhiming Yuan used computer simulations to study the effect of the mother duck's wake.

C. Zhiming Yuan studied the physics behind the fact that by being pushed in a forward direction by waves, ducklings save energy.

D. Naval architect Zhiming Yuan discovered that ducklings are pushed in a forward direction by the waves of their mother's wake, reducing the effect of drag by 158%.

3.38 2398dbd5-d003-4dd5-977c-aabd57c6824a · Hard

While researching a topic, a student has taken the following notes:

- A 2024 study analyzed the facial expressions of wolves and domestic dogs.
- In the study, facial expressions were coded under 46 different facial actions.
- The “ears rotator” facial action is seen in wolves.
- Dog breeds with erect (wolf-like) ears can produce the “ears rotator” facial action.
- Dog breeds with flopped or semi-flopped (non-wolf-like) ears cannot produce the “ears rotator” facial action.

The student wants to compare dog breeds with wolf-like ears to dog breeds with non-wolf-like ears. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. In a 2024 study, dog breeds with non-wolf-like ears were able to produce the same facial movements that wolves could.

B. One difference between dog breeds with wolf-like ears and dog breeds without them is that wolf-like breeds cannot produce the “ears rotator” facial action.

C. Non-wolf-like ears are flopped or semi-flopped, but wolf-like ears are different: they are erect and can produce the “ears rotator” facial action.

D. Like wolves, dog breeds with erect ears can produce the “ears rotator” facial action, while those with flopped or semi-flopped ears cannot.

3.39 ab40fd7d-4d0a-4e63-a40b-8530b5444a6a · Hard

While researching a topic, a student has taken the following notes:

- One of history's greatest libraries was the House of Wisdom in Baghdad, Iraq.
- It was founded in the eighth century with the goal of preserving all the world's knowledge.
- Scholars at the House of Wisdom collected ancient and contemporary texts from Greece, India, and elsewhere and translated them into Arabic.
- Writings included those of the Greek philosopher Aristotle and the Indian mathematician Aryabhata.
- The House of Wisdom used Chinese papermaking technology to create paper versions to be studied and shared.

The student wants to explain how the House of Wisdom preserved the world's knowledge. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. The House of Wisdom was known for bringing together knowledge from around the world, including from Greece, India, and China.

B. Founded in Iraq in the eighth century, the House of Wisdom employed many scholars as translators.

C. Writings from the Greek philosopher Aristotle and the Indian mathematician Aryabhata were preserved at the House of Wisdom.

D. The House of Wisdom collected writings from different countries and created paper versions in Arabic to be studied and shared.

3.40 6b7287f9-e7c2-4415-86f7-d69ac48ef281 · Hard

While researching a topic, a student has taken the following notes:

- Producing the nutrient-rich cyanobacterium *L. maxima* at industrial scale requires high-quality samples of *L. maxima* DNA.
- Yirlis Yadeth Pineda-Rodriguez and a team of researchers at the University of Córdoba, Colombia, evaluated the quantity and purity of *L. maxima* DNA extracted using three different DNA extraction kits.
- CTAB 2X (kit 1) had a DNA yield of 2,134 nanograms per microliter (ng/μL) and a purity ratio of 2.2.
- Pbact (kit 2) had a DNA yield of 157 ng/μL and a purity ratio of 1.6.
- Pplant (kit 3) had a DNA yield of 12.5 ng/μL and a purity ratio of 1.5.
- According to the researchers, Pbact was the most effective because it was the only one with both a sufficiently high yield and a purity rate close to the ideal of 1.8.

The student wants to emphasize the significance of a similarity between two of the kits. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Due to their insufficient yield or purity, CTAB 2X and Pplant were deemed by the researchers to be less effective than Pbact.

B. Compared to CTAB 2X, which had a DNA yield of 2,134 ng/μL, both Pbact and Pplant had insufficient yields; Pplant, in particular, was ineffective due to its low yield.

C. CTAB 2X and Pplant both had a DNA yield above 10 and a purity ratio above 1.4.

D. With the ideal purity ratio being 1.8, CTAB 2X and Pbact were equal in purity, according to the researchers.

3.41 2a121475-3531-4130-9a6f-9b4b2ac0cc54 · Hard

While researching a topic, a student has taken the following notes:

- The *Atlantic Monthly* magazine was first published in 1857.
- The magazine focused on politics, art, and literature.
- In 2019, historian Cathryn Halverson published the book *Faraway Women and the "Atlantic Monthly."*
- Its subject is female authors whose autobiographies appeared in the magazine in the early 1900s.
- One of the authors discussed is Juanita Harrison.

The student wants to introduce Cathryn Halverson's book to an audience already familiar with the *Atlantic Monthly*. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Cathryn Halverson's *Faraway Women and the "Atlantic Monthly"* discusses female authors whose autobiographies appeared in the magazine in the early 1900s.

B. A magazine called the *Atlantic Monthly*, referred to in Cathryn Halverson's book title, was first published in 1857.

C. *Faraway Women and the "Atlantic Monthly"* features contributors to the *Atlantic Monthly*, first published in 1857 as a magazine focusing on politics, art, and literature.

D. An author discussed by Cathryn Halverson is Juanita Harrison, whose autobiography appeared in the *Atlantic Monthly* in the early 1900s.

3.42 514e4a8c-074b-4212-9e5b-95cd02f9d16d · Hard

While researching a topic, a student has taken the following notes:

- The factors that affect clutch size (the number of eggs laid at one time) have been well studied in birds but not in lizards.
- A team led by Shai Meiri of Tel Aviv University investigated which factors influence lizard clutch size.
- Meiri's team obtained clutch-size and habitat data for over 3,900 lizard species and analyzed the data with statistical models.
- Larger clutch size was associated with environments in higher latitudes that have more seasonal change.
- Lizards in higher-latitude environments may lay larger clutches to take advantage of shorter windows of favorable conditions.

The student wants to emphasize the aim of the research study. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Researchers wanted to know which factors influence lizard egg clutch size because such factors have been well studied in birds but not in lizards.

B. After they obtained data for over 3,900 lizard species, researchers determined that larger clutch size was associated with environments in higher latitudes that have more seasonal change.

C. We now know that lizards in higher-latitude environments may lay larger clutches to take advantage of shorter windows of favorable conditions.

D. Researchers obtained clutch-size and habitat data for over 3,900 lizard species and analyzed the data with statistical models.

3.43 47825dd9-caac-44a1-b61d-8a36623c6ece · Hard

While researching a topic, a student has taken the following notes:

- The National Congress of American Indians (NCAI) was founded in 1944 by representatives of fifty tribal governments.
- The NCAI was created to protect the sovereignty of Indigenous tribes.
- Napoleon B. Johnson (Cherokee) was the NCAI's first president.
- In 1975, the US Congress passed the Indian Self-Determination and Education Assistance Act (Public Law 96-638).
- This legislation formally acknowledged tribes' right to self-governance.
- The advocacy of the NCAI was a key factor in the law's passing.

The student wants to identify an accomplishment of the NCAI. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. The NCAI, founded by representatives of fifty tribal governments, had Napoleon B. Johnson (Cherokee) as its first president.

B. Founded in 1944, the NCAI was created by representatives of tribal governments from fifty sovereign Indigenous tribes.

C. The NCAI's advocacy was key to the passing of Public Law 96-638, legislation formally acknowledging Indigenous tribes' right to self-governance.

D. In 1975, the NCAI passed the Indian Self-Determination and Education Assistance Act, which was created to protect the sovereignty of Indigenous tribes.

3.44 84f5f227-03be-40cf-addc-599ce1a63da9 · Hard

While researching a topic, a student has taken the following notes:

- Cambodia's Angkor Wat was built in the 1100s to honor the Hindu god Vishnu.
- It has been a Buddhist temple since the sixteenth century.
- Decorrelation stretch analysis is a novel digital imaging technique that enhances the contrast between colors in a photograph.
- Archaeologist Noel Hidalgo Tan applied decorrelation stretch analysis to photographs he had taken of Angkor Wat's plaster walls.
- Tan's analysis revealed hundreds of images unknown to researchers.

The student wants to present Tan's research to an audience unfamiliar with Angkor Wat. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Tan photographed Angkor Wat's plaster walls and then applied decorrelation stretch analysis to the photographs.

B. Decorrelation stretch analysis is a novel digital imaging technique that Tan used to enhance the contrast between colors in a photograph.

C. Using a novel digital imaging technique, Tan revealed hundreds of images hidden on the walls of Angkor Wat, a Cambodian temple.

D. Built to honor a Hindu god before becoming a Buddhist temple, Cambodia's Angkor Wat concealed hundreds of images on its plaster walls.

3.45 9ed92caf-8ee3-4252-b86b-47b5bce47519 · Hard

While researching a topic, a student has taken the following notes:

- Modularity of mind is the notion that the mind is at least partly composed of innate neural structures (modules) that perform fast, necessary tasks.
- 1983: cognitive scientist Jerry A. Fodor hypothesized that low-level cognitive systems (e.g., perception, language) are modular.
- In Fodorian modularity, high-level systems (e.g., reasoning) are not modular.
- 2003: cognitive scientist Peter Carruthers proposed the massive modularity hypothesis (MMH).
- MMH expands modularity to include all cognitive systems.

The student wants to compare Fodor's hypothesis with Carruthers's. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. In considering some but not all cognitive systems modular, Fodorian modularity is not as expansive in its definition of modularity as MMH is.

B. Following Fodor's 1983 hypothesis, Carruthers proposed that modularity of mind includes all cognitive systems.

C. The hypotheses of Fodor and Carruthers differ in whether they consider low-level cognitive systems, such as perception and language, modular.

D. In 2003, Carruthers proposed the massive modularity hypothesis, disagreeing with Fodor's earlier hypothesis that the mind is composed of innate neural structures.

3.46 57bc16b7-b31e-43cd-9f03-a9aea8afed33 · Hard

While researching a topic, a student has taken the following notes:

- In 1978, Sámi activists staged protests to block the construction of a dam on the Alta River in Norway.
- The dam would disrupt Sámi fishing and reindeer herding.
- The dam was ultimately built, but the Alta conflict had a lasting impact.
- It brought international attention to the issue of Sámi rights.
- It led to a set of 2005 legal protections establishing Sámi rights to lands, waters, and resources.

The student wants to make and support a generalization about the Alta conflict. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. During the Alta conflict, Sámi activists staged protests to block the construction of a dam on the Alta River in Norway that would disrupt local fishing and reindeer herding.

B. Although the dam that the Sámi activists had protested was ultimately built, the Alta conflict had a lasting impact.

C. Sámi rights to lands, waters, and resources received international attention and legal protections as a result of the Alta conflict.

D. The Alta conflict had a lasting impact, resulting in international attention and legal protections for Sámi rights to lands, waters, and resources.

3.47 abe6747a-221f-4f61-bcd7-d10dd928c01b · Hard

While researching a topic, a student has taken the following notes:

- Two opposing theories of vision divided scholars for many centuries.
- The ancient Greek mathematician Euclid (circa 300 BCE) supported the extramission theory.
- This theory held that the eyes emit a form of radiation that illuminates objects in its range.
- The ancient Greek philosopher Aristotle (384–322 BCE) supported the intromission theory.
- This theory held that objects emit a form of radiation that reaches the eyes.
- In the eleventh century, Arab mathematician Ibn al-Haytham (965–1040 CE) largely settled the debate with the first conclusive experiments supporting intromission.

The student wants to provide a historical overview of the two theories. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Scholars were divided between the extramission and intromission theories of vision until Ibn al-Haytham's eleventh-century experiments largely settled the debate in support of intromission.

B. Through two opposing theories of vision—extramission and intromission—Euclid, Aristotle, and Ibn al-Haytham held that a form of radiation is emitted either from objects or from the eyes.

C. While Ibn al-Haytham largely settled the debate in the eleventh century, Aristotle supported the theory of intromission centuries before.

D. Before the eleventh century, the ancient Greek philosopher Aristotle supported the intromission theory, which held that objects emit a form of radiation that reaches the eyes.

3.48 6b2053d2-4dcb-49c5-a3a1-70df096d9d56 · Hard

While researching a topic, a student has taken the following notes:

- A commodity chain is the series of links connecting the production and purchase of a commodity on the world market.
- Chinese American anthropologist Anna Tsing studies the contemporary commodity chain of matsutake mushrooms.
- At one end of the matsutake chain are mushroom pickers in Oregon.
- At the other end are wealthy consumers who buy the costly matsutake in Japan.
- According to Tsing, “Japanese traders began importing matsutake in the 1980s, when the scarcity of matsutake in Japan first became clear.”

The student wants to provide an overview of the matsutake commodity chain. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. The contemporary matsutake commodity chain has its origins in the 1980s when, according to Tsing, “the scarcity of matsutake in Japan first became clear.”

B. Commodity chains include the linked production and purchase of commodities, such as the matsutake mushroom, on the world market.

C. Decades after the Japanese import of matsutake began, a commodity chain now links matsutake pickers in Oregon with wealthy consumers of the costly mushrooms in Japan.

D. Wealthy consumers who buy the costly mushrooms in Japan are at one end of the matsutake commodity chain.

3.49 d01857e1-22bc-493b-b6d0-dd1a3127bdaf · Hard

While researching a topic, a student has taken the following notes:

- Neuroscientists Krishnan Padmanabhan and Zhen Chen sought to better understand the workings of the brain's olfactory system.
- They devised a study using mathematical models.
- They found that certain fibers allow the brain to toggle from one method of processing smells to another.
- In one method, cells in the piriform cortex (where the perception of odor forms) capture olfactory information at a given moment.
- In the other, the cells track changes in olfactory information over time.

The student wants to summarize the study's findings. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. To arrive at these findings, which describe dual methods of processing smells in the piriform cortex, Padmanabhan and Chen devised a study using mathematical models.

B. Padmanabhan and Chen showed that olfactory information is captured by cells in the piriform cortex, where the perception of odor forms.

C. Using mathematical models, Padmanabhan and Chen devised a study to better understand the workings of the brain's olfactory system.

D. According to Padmanabhan and Chen, the brain can toggle between capturing olfactory information at a given moment and tracking changes in that information over time.

3.50 d49db9eb-fd01-4f5e-8520-79aad46ad0c7 · Hard

While researching a topic, a student has taken the following notes:

- In a 2023 study, environmental scientist Jazmin Locke-Rodriguez and colleagues tested the use of floating treatment wetlands (FTWs) in Florida.
- FTWs are artificial floating platforms of plants used to remediate polluted or nutrient-imbalanced water.
- Finding: FTWs using marigold flowers removed 52% more total phosphorus than the control.
- Finding: The test yielded 65 market-quality blooms per square meter.
- The authors concluded marigolds showed “promising potential as a commercially viable remediating crop cultivated on FTWs in South Florida.”

The student wants to present the findings of the study. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. The authors concluded that marigolds grown on FTWs were “commercially viable,” having produced 65 blooms per square meter of market-quality blooms in a 2023 study.

B. In a 2023 study, Locke-Rodriguez and colleagues found that marigolds cultivated on FTWs produced 52% more market-quality flower blooms than the control.

C. Locke-Rodriguez and colleagues found that FTWs using marigolds not only helped remove phosphorus from the water but also yielded market-quality blooms.

D. FTWs using marigolds, Locke-Rodriguez and colleagues found, yielded 65 flower blooms and removed 52% of phosphorus from the water.

3.51 e4051f7d-72b1-4839-919f-d942b9f73ba4 · Hard

While researching a topic, a student has taken the following notes:

- A small number of US Navy sailors of Filipino descent served during the US Civil War (1861–1865).
- Stephen Amos was born in the Philippines around 1830.
- He enlisted in the US Navy in November 1863.
- Raphael Ignases was born in the Philippines around 1834.
- He enlisted in the US Navy in July 1861.

The student wants to emphasize the historical significance of Stephen Amos's enlistment date. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Both Stephen Amos and Raphael Ignases were US Navy sailors of Filipino descent, but Amos enlisted in the Navy in 1863, two years later than Ignases.

B. Stephen Amos was a US Navy sailor of Filipino descent, along with Raphael Ignases, who was born in the Philippines around 1834.

C. Stephen Amos enlisted in the US Navy in 1863, making him one of the few sailors of Filipino descent to serve in the US Civil War (1861–1865).

D. When Stephen Amos enlisted in the US Navy in November 1863, he joined sailors such as Raphael Ignases, who had been born in the Philippines around 1834.

3.52 c93c8f6e-0b3e-4c5a-b751-a2d724dec205 · Hard

While researching a topic, a student has taken the following notes:

- The magnificent frigatebird (*fregata magnificens*) is a species of seabird that feeds mainly on fish, tuna, squid, and other small sea animals.
- It is unusual among seabirds in that it doesn't dive into the water for prey.
- One way it acquires food is by using its hook-tipped bill to snatch prey from the surface of the water.
- Another way it acquires food is by taking it from weaker birds by force.
- This behavior is known as kleptoparasitism.

The student wants to emphasize a similarity between the two ways a magnificent frigatebird acquires food. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. A magnificent frigatebird never dives into the water, instead using its hook-tipped bill to snatch prey from the surface.

B. Neither of a magnificent frigatebird's two ways of acquiring food requires the bird to dive into the water.

C. Of the magnificent frigatebird's two ways of acquiring food, only one is known as kleptoparasitism.

D. In addition to snatching prey from the water with its hook-tipped bill, a magnificent frigatebird takes food from other birds by force.

3.53 5cc70221-e8d0-4917-95ba-4d9ec35d842b · Hard

While researching a topic, a student has taken the following notes:

- In North America, woodlands have expanded into areas that were once grasslands.
- Thomas Rogers and F. Leland Russell of Wichita State University investigated whether woodland expansion is related to changes in climate.
- Rogers and Russell analyzed core samples from oak trees on a site that was not wooded in the past and indexed the age of the trees with historical climate data to see if tree populations and climate were correlated.
- Tree population growth was associated with dry intervals.
- Droughts may have played a role in woodland expansion.

The student wants to emphasize the aim of the research study. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Thomas Rogers and F. Leland Russell, researchers at Wichita State University, wanted to know if woodland expansion is related to changes in climate.

B. Thanks to the work done by Thomas Rogers and F. Leland Russell, we now know that droughts may have played a role in woodland expansion.

C. Wichita State University researchers have determined that tree population growth was associated with dry intervals.

D. Thomas Rogers and F. Leland Russell analyzed core samples from oak trees on a site that was not wooded in the past, indexing the age of the trees with historical climate data.

3.54 38a5c95b-f2c6-4f5c-95c4-9054cd57ded1 · Hard

While researching a topic, a student has taken the following notes:

- Malapportionment is the over- or underrepresentation (relative to population size) of electoral districts in a governing body.
- It is a common feature of representative governments.
- There are 169 seats in Norway's supreme legislature (the Storting).
- Seats are distributed by a formula that awards 1 point per resident and 1.8 points per unit of land.
- Less populated rural districts with large tracts of land receive a disproportionate number of seats compared to smaller but more populated urban districts.

The student wants to refute a claim that malapportionment in the Storting favors small urban districts. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Less populated rural districts are disproportionally underrepresented in the Storting, creating an unfair advantage for smaller but more populated urban districts.

B. It's untrue that malapportionment in the 169-seat Storting favors small urban districts; rather, the formula for distributing seats overrepresents more populated districts.

C. A common feature of representative governments, malapportionment occurs when electoral districts are over- or underrepresented.

D. Awarding more points per unit of land than points per resident, the formula for distributing Storting seats overrepresents less populated rural districts with large tracts of land.

3.55 48c6e022-9241-4853-b508-7857271ddca7 · Hard

While researching a topic, a student has taken the following notes:

- Freddie Wong (born 1985) is a director and special effects artist from the United States.
- He is best known for the action-comedy web series *Video Game High School* (*VGHS*).
- *VGHS* premiered in 2012 on RocketJump, a YouTube channel that Wong cocreated.
- The series was celebrated for its inventive video game–centric world and high-quality special effects.
- *VGHS* was nominated for a Producers Guild Award for Outstanding Digital Series.

The student wants to begin a narrative about Wong's award-nominated web series. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. In 2012, director and visual effects artist Freddie Wong launched a new action-comedy web series: *Video Game High School.*

B. *Video Game High School* was celebrated for its inventive video game–centric world and high-quality special effects, and it was nominated for a Producer's Guild Award for Outstanding Digital Series.

C. Wong, cocreator of the YouTube channel RocketJump, would go on to see his web series be nominated for a Producers Guild Award.

D. In 2012, *Video Game High School* premiered on RocketJump; it would later be nominated for an award.

3.56 006d3b30-cede-4889-82fd-dbf2335054f4 · Hard

While researching a topic, a student has taken the following notes:

- Digital Light Synthesis (DLS) is a form of additive manufacturing that utilizes light to rapidly cure liquid resin into high-quality, 3D objects.
- Step 1: Ultraviolet (UV) light images are projected up into a pool of liquid resin, where the object's first layer takes shape.
- Step 2: The partially cured resin object is raised, leaving a thin space (a "dead zone") beneath it for oxygen and liquid resin to flow through.
- Step 3: The UV light passes through the dead zone—maintaining the flow of resin—and partially cures additional layers of the object.
- Step 4: When the resin object is complete, it is baked in an oven to complete the curing.

The student wants to describe how DLS cures 3D objects. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. DLS is a form of additive manufacturing that creates a "dead zone" in which UV light solidifies layer by layer before being baked in an oven, creating a high-quality, 3D object.

B. DLS cures 3D objects by passing through a "dead zone," adding layers to the object, then curing the object in an oven.

C. In DLS, UV light images are projected into a liquid resin pool to cure a 3D object layer by layer; once solidified, the object is baked in an oven.

D. In DLS, UV light is projected into layers of liquid resin until the resin solidifies and passes through a "dead zone," wherein the curing is completed.

3.57 e82e8e87-07e4-4651-aa1e-23f2fa389d6e · Hard

While researching a topic, a student has taken the following notes:

- Researchers in a 2021 study wanted to determine the rate at which 17 languages conveyed both information and syllables.
- They calculated the bits of information conveyed per second (the IR, or information rate).
- The IR was found to be approximately consistent across the 17 languages (an average of 39 bits per second).
- They calculated the number of syllables spoken per second (the SR, or syllable rate).
- Spanish had the second-fastest SR (7.7 syllables per second).
- Vietnamese had the sixteenth-fastest SR (5.3 syllables per second).

The student wants to present an overview of the study's findings. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. The 2021 study determined the information rate (IR) of 17 languages in bits of information conveyed per second.

B. Researchers found that information was conveyed more quickly in Spanish, at 7.7 syllables per second, than in Vietnamese, at 5.3 syllables per second.

C. Vietnamese had the sixteenth-fastest syllable rate, lower than that of Spanish, which had the second-fastest; however, Spanish had the lower information rate of the two.

D. Though some of the languages differed in number of syllables spoken per second, all 17 conveyed information at roughly the same rate.

3.58 f44c0976-8eea-4f60-9365-e7684274c4d9 · Hard

While researching a topic, a student has taken the following notes:

- Phobetor, a name drawn from Greek mythology, is an exoplanet that orbits the star PSR B1257+12, also known as Lich.
- Phobetor's mass is 0.01 times that of Jupiter, or 0.01 Jupiter masses.
- Mastika, which means "gem" or "jewel" in Malay, is an exoplanet that orbits the star HD 179949, also known as Gumala.
- Mastika's mass is 0.92 Jupiter masses.

The student wants to make and support a generalization about exoplanets. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Exoplanets that are named Phobetor orbit Lich, and those that are named Mastika orbit Gumala.

B. Even though Phobetor and Mastika are both exoplanets, their masses are different: Phobetor's mass is 0.01 Jupiter masses, and Mastika's is 0.92 Jupiter masses.

C. Many stars have both a designation and a proper name; for instance, PSR B1257+12 is also known as Lich, and HD 179949 is also known as Gumala.

D. Exoplanet names have diverse origins, a fact that can be seen in the cases of Phobetor, a name drawn from Greek mythology, and Mastika, which means "gem" or "jewel" in Malay.

3.59 42442008-1f03-4709-8ee5-daab0e253184 · Hard

While researching a topic, a student has taken the following notes:

- Suzanne K. Birner led a study analyzing rocks on the seafloor to better understand the history of Earth's mantle.
- Rock samples were collected from two seafloor ridges.
- The researchers determined the samples' period of formation (the Archean eon) and oxidation level (extremely low).
- High temperatures in the Archean likely caused the rocks' low oxidation.
- Birner's team suggests the oxidation of Earth's mantle has remained stable over time, contrary to previous theories.
- The findings help explain the unique conditions that allowed life to develop on Earth.

The student wants to present the study's research methods. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Birner led a study to better understand the history of Earth's mantle and explain the conditions that allowed life to develop.

B. To further analyze the origins of Earth's unique conditions, researchers focused on rocks from the Archean eon, when Earth's temperatures were extremely high.

C. By studying these ancient rocks, the team aimed to challenge previous theories about changes in Earth's mantle over time.

D. Birner's team analyzed the age and oxidation levels of rock samples collected from two seafloor ridges.

3.60 895349d7-a567-400f-9de6-a4f1de149e22 · Hard

While researching a topic, a student has taken the following notes:

- African American women played prominent roles in the Civil Rights Movement, including at the famous 1963 March on Washington.
- Civil rights activist Anna Hedgeman, one of the march's organizers, was a political adviser who had worked for President Truman.
- Civil rights activist Daisy Bates was a well-known journalist and advocate for school desegregation.
- Hedgeman worked behind the scenes to make sure a woman was included in the lineup of speakers at the march.
- Bates was the sole woman to speak, delivering a brief but memorable address to the cheering crowd.

The student wants to compare the two women's contributions to the March on Washington. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Hedgeman and Bates contributed to the march in different ways; Bates, for example, delivered a brief but memorable address.

B. Hedgeman worked in politics and helped organize the march, while Bates was a journalist and school desegregation advocate.

C. Although Hedgeman worked behind the scenes to make sure a woman speaker was included, Bates was the sole woman to speak at the march.

D. Many African American women, including Bates and Hedgeman, fought for civil rights, but only one spoke at the march.

3.61 020f2cdc-bab1-476b-849b-762e11ace6c1 · Hard

While researching a topic, a student has taken the following notes:

- Leigh Torres is a marine ecologist.
- She conducted a study of blue whales in New Zealand's South Taranaki Bight region.
- She wanted to know how ocean temperature affects where the whales forage for krill in that region.
- She found that during a marine heat wave, the whales foraged farther offshore than they had during cooler periods.
- The offshore waters, which were colder than areas closer to shore, had a higher relative abundance of krill.

The student wants to emphasize the aim of the research study. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Analyzing ocean temperature data, Torres found that during a marine heat wave, blue whales foraged farther offshore than they had during cooler periods.

B. In her study, Torres sought to determine how ocean temperature affects where blue whales forage for krill in the South Taranaki Bight region.

C. Torres's study revealed that blue whales were attracted to offshore waters with a relatively high abundance of krill.

D. Torres, a marine ecologist, studied blue whales in the South Taranaki Bight region, where the whales forage.

3.62 3b0557c9-cb2e-4757-ac8b-73a2f43694cb · Hard

While researching a topic, a student has taken the following notes:

- Marcela Guerrero is a curator at the Whitney Museum of American Art in New York.
- She curated the Whitney's 2018 exhibition *Pacha, Llaqta, Wasichay: Indigenous Space, Modern Architecture, New Art*.
- This exhibition featured works by seven emerging Latino artists.
- She curated the Whitney's 2020 exhibition *Vida Americana: Mexican Muralists Remake American Art, 1925–1945*.
- This exhibition included nearly 200 works by twentieth-century Latino and Mexican artists.

The student wants to describe the exhibition that Guerrero curated in 2018. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Held at the Whitney Museum of American Art, the exhibition *Vida Americana: Mexican Muralists Remake American Art, 1925–1945* included nearly 200 works by twentieth-century Mexican and Latino artists.

B. *Pacha, Llaqta, Wasichay: Indigenous Space, Modern Architecture, New Art*, an exhibition at the Whitney Museum of American Art, featured works by seven emerging Latino artists.

C. In both 2018 and 2020, Marcela Guerrero curated exhibitions at the Whitney Museum of American Art in New York.

D. While one exhibition that Marcela Guerrero curated featured works by emerging artists, another included works by twentieth-century artists.

3.63 f5cdc738-f25b-4daa-8a0a-e4a24333ca68 · Hard

While researching a topic, a student has taken the following notes:

- Shanawdithit (1801–1829) was a Beothuk cartographer (mapmaker).
- Her maps of Newfoundland's Beothuk Lake outline both the lake and various points around the lake where encounters between the Indigenous Beothuk people and British colonists occurred.
- Her maps are notable for depicting the experiences the Beothuk had within the landscape.
- Contemporary Potawatomi cartographer Margaret Pearce: Indigenous cartography emphasizes "experienced space, or place, as opposed to the Western convention of depicting space as universal, homogenized, and devoid of human experience."
- Pearce: "Indigenous cartographies are as diverse as Indigenous cultures, from Hawaiian performative cartographies to Navajo verbal maps and sand paintings."

The student wants to describe Shanawdithit's approach and explain its significance. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Shanawdithit's maps are part of a broader tradition of Indigenous cartography that, according to Pearce, ranges from "Hawaiian performative cartographies to Navajo verbal maps and sand paintings."

B. Shanawdithit mapped Beothuk Lake through significant encounters that occurred there, an approach described as "depicting space as universal [and] homogenized."

C. According to Pearce, Indigenous cartography, such as Shanawdithit's maps of Beothuk Lake, emphasizes "experienced space, or place," with a variety of approaches that reflect the diversity of Indigenous cultures.

D. By depicting experiences of the Beothuk that occurred around Beothuk Lake, Shanawdithit's maps reflect Indigenous cartography's emphasis on "experienced space, or place" rather than the landscape alone.

3.64 7ba8f9bf-566c-4c93-88c3-ef072c125b0c · Hard

While researching a topic, a student has taken the following notes:

- The background colors of US and UK road signs are used to denote each sign's purpose.
- Directional signs are a type of sign containing information such as route names, distance to a destination, etc.
- Highways, major roadways, and minor roadways in the US generally use green for directional signs.
- Highways in the UK generally use blue for directional signs.
- Major roadways in the UK generally use green for directional signs.

The student wants to contrast how green backgrounds are used in US and UK road signs. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Blue backgrounds are generally used on directional highway signs in the UK, in contrast to the green highway signs used in the US.

B. In the US, green signs indicate directional information on both highways and major roadways, but in the UK, directional signs of this color generally appear only on major roadways.

C. Green road signs are used in both the US and UK to denote directional information on major roadways, like distance to a destination or route names.

D. Both the UK and the US use directional signs, which include information on route names and distance to a destination.

3.65 122def51-3b7a-48b3-80ed-7c47c5fd5ec5 · Hard

While researching a topic, a student has taken the following notes:

- Bike-share programs provide bicycles for shared use.
- In docked bike sharing, riders rent a bike and return it to designated docking stations.
- Docked programs are orderly and offer consistency to riders but require significant space and money to implement.
- In dockless bike sharing, riders locate a bike and leave it wherever they choose.
- Dockless programs are relatively simple and inexpensive to implement and offer flexibility to riders.
- Dockless programs can be disorganized.

The student wants to compare some disadvantages of docked and dockless bike-share programs. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Dockless programs can be disorganized; docked programs, on the other hand, offer order and consistency.

B. Worth noting is that while dockless programs are relatively easy and inexpensive to implement, they are less flexible than docked programs.

C. Docked programs are more resource-intensive than dockless programs, but they avoid some of the latter's organizational challenges.

D. Though dockless programs offer flexibility, docked bike-share programs provide bicycles for shared use.

3.66 944d6e30-8dda-4769-bf70-e1b924a62c8c · Hard

While researching a topic, a student has taken the following notes:

- Political scientist Graham Allison is known for his Thucydides trap theory.
- Allison's theory states that whenever "a rising power is threatening to displace a ruling power," conflict is likely.
- The theory is based on Thucydides's explanation of the conflict between Athens and Sparta.
- Thucydides wrote that "the rise of Athens and the fear this instilled in Sparta" made conflict "inevitable."
- History professor Edmund Stewart recently challenged the historical basis of the theory.
- Stewart claimed that Athens was not a rising power and that the rivals experienced a "clash of cultures" instead.

The student wants to use a quotation to challenge Thucydides's explanation of the conflict between Athens and Sparta. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. According to Allison's Thucydides trap theory, whenever "a rising power is threatening to displace a ruling power," conflict is likely.

B. Thucydides wrote that conflict between the two powers was "inevitable," although Stewart later challenged the historical basis of this claim.

C. According to Stewart, a "clash of cultures" between Athens and Sparta caused the conflict, not Athens's rise.

D. Thucydides explained that conflict was caused by "the rise of Athens and the fear this instilled in Sparta," but Allison disagreed, seeing the conflict as an example of the Thucydides trap.

3.67 7dca01cd-5a38-4868-b396-5f2e3975d2e7 · Hard

While researching a topic, a student has taken the following notes:

- A sestina is a thirty-nine-line poetic form.
- Each line of the poem ends with one of six end words, which alternate according to a set pattern.
- “Forage Sestina” is a sestina by Marilyn Hacker.
- Its end words are *words*, *structure*, *wire*, *beam*, *wall*, and *room*.
- “Towards Autumn” is a sestina by Marilyn Hacker.
- Its end words are *daughter*, *friend*, *bread*, *mother*, *lover*, and *myself*.

The student wants to use one of the poems to illustrate the sestina form. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Hacker employs the sestina, a poetic form with thirty-nine lines and six end words, in both “Forage Sestina” and “Towards Autumn.”

B. As a sestina, “Towards Autumn” contains thirty-nine lines and six end words—in this case, *daughter*, *friend*, *bread*, *mother*, *lover*, and *myself*—that alternate in a set pattern.

C. The thirty-nine-line sestina form uses the words *daughter*, *friend*, *bread*, *mother*, *lover*, and *myself*, which are found in the poem “Forage Sestina.”

D. Hacker has used the sestina form multiple times, as in “Towards Autumn,” which contains these six words: *words*, *structure*, *wire*, *beam*, *wall*, and *room*.

3.68 cf3bbc21-218a-40aa-a02c-2249bcd22e0a · Hard

While researching a topic, a student has taken the following notes:

- The nautical mile (6,076 feet) is the measure of distance used in seafaring navigation.
- A nautical mile directly correlates to one minute (1/60th of a degree) of latitude.
- The curvature of Earth affects the accurate measurement of long distances when using flat maps.
- Measuring distances with latitude and longitude coordinates takes into account Earth's curvature.
- Mariners use nautical charts marked with latitude and longitude to quickly calculate distances and positions.

The student wants to explain why nautical miles are used to measure distances in seafaring navigation. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Nautical miles are a measure of distance equal to one minute of latitude, which is a feature nautical charts use to calculate distances and positions.

B. Since they directly correlate to the coordinates on nautical charts, which take into account Earth's curvature, nautical miles are an efficient way to calculate distances at sea.

C. Using nautical miles for navigation at sea takes Earth's curvature into account, whereas measuring distances with latitude and longitude coordinates does not.

D. Nautical charts use latitude and longitude to measure long distances; these charts are more accurate than flat maps for measuring distances in seafaring navigation because they account for Earth's curvature.